CSE 586 Bütünleme Exam

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I. Question (15 points)

Write the algorithm that handles the boolean query x \ y (difference) using inverted index.

II. Question (12 points)

Let say you have the following terms in the dictionary: adana, istanbul, ankara, izmir, erzurum, tekirdag, hakkari. In order to get

a.) a balanced
b.) a maximumly unbalanced

binary tree, in what order would you insert those terms to the dictionary?

III. Question (15 points)

Consider the following scenarios for unranked evaluation of document retrieval:

a.) There are 10 relevant documents for the given query in the collection, system A returns 5 relevant and 5 irrelevant documents.
b.) There are 10 relevant documents for the given query in the collection, system B returns 10 relevant and 90 irrelevant documents.
c.) There are 10 relevant documents for the given query in the collection, system C returns 4 relevant documents.

Order the systems A, B, and C in terms of precision, recall and F-measure.

IV. Question (14 points)

Suppose that the task is to cluster the following eight points (with (x) representing location) into three clusters. A_1(-2), A_2(3), A_3(8), A_4(5), A_5(14), A_6(16), A_7(12), A_8(9). The distance function is Euclidian distance. Suppose initially we assign A_1, A_4, and A_7 as the center of each cluster, respectively. Run the k-means clustering one iteration on these data.

V. Question (16 points)

docID	words in document	in Europe?
1	Turkey England Greece	yes
2	Russia England Japan	yes
3	Turkey England China	yes
4	Japan China	no
5	Japan China Turkey	no
6	China England India	no

Based on the data given above,

a.) Estimate a multinomial Naive Bayes classifier
b.) Apply classifier to the document
`Turkey England Japan Japan`

VI. Question (14 points)

Based on the data given above

a.) Implement Rocchio classifier, that is find the mean vector of each class.
b.) Apply classifier to the document
`Turkey England Japan Japan`

VII. Question (14 points)

Based on the data given above, estimate the class of the document

`Turkey England Japan Japan`

for

a.) $k = 1$
b.) $k = 3$